Project Proposal: Powder surface analysis using point cloud 3D scanning data

Lukas Vaartjes
Student number: 7046650

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Supervisors: Prof. dr. S. Abeln, Dr. J.H.J. Quodbach, W. Kooijman
Examiner: Dr. W.J. dos Santos Silva

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3D Three Dimensional

AM Additive Manufacturing

API Active Pharmaceutical Ingredient

BJ Binder Jetting

BQ Ball Query

CNN Convolutional Neural Network

FDM Fused Deposition Modeling

FOV Field-of-view

FPS Farthest Point Sampling

HDR High Dynamic Range

Ra Arithmetic mean roughness

Rq Root mean square roughness

SA Set Abstraction

SLA Stereolithography

SLM Selective Laser Melting

SLS Selective Laser Sintering

SSE Semi-Solid extrusion

1 Introduction

While the traditional manufacturing of pharmaceutical tablets is done in a standardized mass-production manner to increase efficiency and decrease costs, it does not take into account the individual needs of patients. People require different doses according to their medical conditions, lifestyles and pharmacogenomics. Personalized medication solves these challenges by tailoring the tablets to the patient. This can prevent side effects and improve the effectiveness of treatment by providing personalized medication based on a person's genetic.[1] Previous research showed that 30% fewer serious side effects occurred when using the genetic information of patients.[1] It was also found that, for example, patients with diabetes[2] benefit from personalized medication because of the increased effectiveness.[3,4]

Because traditional manufacturing is so efficient, producing personalized tablets with these machines is expensive and labor-intensive.[5,6] Three-dimensional **(3D)** printing, can be used for personalized drug production while maintaining high efficiency and low costs.[7] 3D printing consists of multiple types of Additive Manufacturing **(AM)** techniques and enables the manufacturing of structures and geometries from model data that is represented in a three-dimensional space. For the production of pharmaceutical tablets, the most common extrusion-based[8] manufacturing methods are Semi-Solid Extrusion[9] **(SSE)** and Fusion Deposition Modeling[8] **(FDM)**, while the most common powder-based[10] manufacturing methods are Selective Laser Sintering[11] **(SLS)**, Binder Jetting[12] **(BJ)**. Powder-based 3D printing has the highest resemblance to the traditional tablet manufacturing process.[7] FDM and SSE require less expensive hardware compared to BJ and SLS, but suffer from low efficiency, as well as low drug loading with FDM.[13]

BJ and SLS both offer high drug loading, fast production and a wide variety of excipients,[13] but the equipment is more expensive. FDM and SSE can be used when low efficiency is not an issue, in a localized setting such as hospitals or community pharmacies. Low efficiency will not work for medication for diabetes because mass production of these personalized tablets is required to meet the demand. BJ and SLS are a valid option to balance the production of tablets with different dosages, without the need to customize tablets for each individual patient. In this case, BJ and SLS are a valid options to achieve this personalization,
BJ uses a binding agent[12] to bind the material together, whereas in SLS a laser is used to solidify the material. A powder mixture is used consisting of the Active Pharmaceutical Ingredient **(API)** and several excipients. Alternating, a layer of powder is spread on a build plate, and the laser or binding agent is used to transfer the powder into a solid form. This process is repeated until the tablets are completed. Detecting defects in an early stage during production is important to get SLS or BJ as efficient as possible and minimize defects in the tablet. The quality of the powder layer is one of the factors that directly impacts the final tablet quality.[14,15] If the powder layer contains defects, the geometric properties, mechanical properties and surface quality will be negatively affected. This results in a weak tablet that falls apart easily, or the dissolving behavior of the tablet changes.[14] As the prediction of how the tablet dissolves will change, the release of the APIs will become unpredictable.

Detecting defects in the powder layer is currently done by means of visual inspection.[16] Larger defects such as holes or cracks are easy to detect with the naked eye, but this will become inconvenient when microcracks, micro holes, laser non-uniformity and layer roughness occur. Detecting these defects is important since they have a big impact on the tablet quality.[16] The analysis of the powder bed in SLS printing has been studied by previous research focusing on defect detection and surface quality of the powder layer. Zhao et al.[17] proposed a real-time powder bed defect detection method, using a camera to generate point clouds for metal powders, Knaak et al.[18] proposed a similar study for metal powders, but using HDR images to do analysis. Another related study is the proposed method of Le et al.[19] for bed defect detection for metal powder using a scanner and determining recoating strategies to remove these defects. Ruther et al.[11] proposed a defect detection using images for polymers such as PA12, and Lupo et al.[20] also used polymers to do powder bed defect detection using a microscopic setup. These studies focussed mostly on powder bed defect detection using polymers or metal powder, whereas this study will use a different kind of material set consisting of different sugars. Instead of using a camera or sensor, this study uses a 3D scanner to generate point cloud data. In this study, an algorithm will be designed to automate the process of defect detection of the powder layer. A 3D scanner is installed on the setup with a printbed and blade recoater to simulate the spreading of powder by a SLS or BJ printer. The 3D scanner will gather the necessary data for an algorithm to analyze and classify defects in the powder layer. This can be applied to both SLS and BJ printing as they use similar powder spreading and build plate mechanism. However, this study will focus only on SLS printing as only a SLS machine was available.

2 Description of SLS printing process

Selective Laser Sintering **(SLS)** is an additive manufacturing process and uses a laser to fuse powdered material together and build a 3D object layer by layer. The build plate is where the model is manufactured and the powder reservoir stores the material used during printing. The model is created layer by layer. For each new layer, the build plate is lowered by the layer height parameter, and the powder reservoir is raised by the amount of material necessary for a single layer. To make sure that sufficient material is available for the construction of a layer, slightly more material is provided than required. The material for a layer is distributed by a recoater, either a blade or roller, and moves across the build plate to spread the material into an even layer. When a new powder layer is spread across the build plate, the laser, responsible for the sintering process, shoots at specific X and Y coordinates on the powder layer.[6,7] The laser is directed using a mirror galvanometer and focused with an F-theta lens.

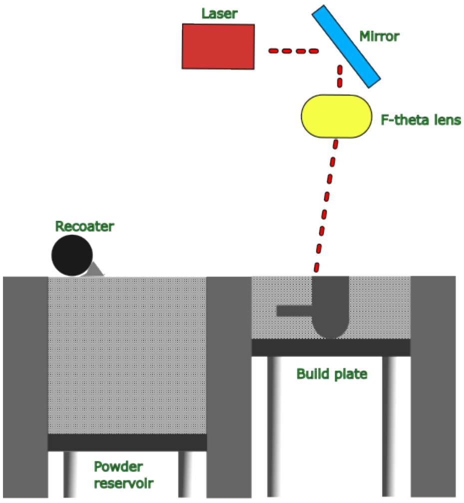

Figure 1: Schematic representation of a simplified SLS printer

2.1 Parameters for SLS printing

Defects during printing are related to the type of material and print parameters.[21] Optimized materials and parameters are required to improve the result by minimizing defects and maximizing efficiency. The three main components that can be adjusted with parameters are the roller, the powder bed and the laser.[21] The laser can be adjusted by the laser power and scanning speed parameters. The density, hardness and surface roughness[15] of the tablets are greatly influenced by these parameters.[22] The powder bed can be adjusted by its temperature and the layer height parameters. A small layer height will result in a smoother surface finish. However, the layer height should not be smaller than the particle size because of the recoater. The bed temperature limits the stress on the object and reduces the chance of deformation. The translational velocity of the recoating mechanism can be adjusted. In the case of the roller as the recoating mechanism, the rotational speed can be adjusted as well. The density of the powder bed, determined by the speed of the recoater, influences the mechanical properties and other pharmaceutical properties of the tablet such as **(API)** content. Weak mechanical properties of the tablet are also caused by defects in the powder layer on the printbed and reduces the overall strength of the tablet.[15] The density and defects influence the drug release, changing the behavior of the disintegration.[14] More information about common defects and their influence on the tablets is discussed in section 2.2.

2.2 Powder layer and flowability defects

Defects in the powder layer can occur during printing and are related to the type of material and print parameters. If the spreading behavior is not uniform, inconsistencies and defects will occur in every layer, reducing the layer bonding between layers. This will result in a tablet that has low mechanical strength, falls apart and dissolves too fast.[14] For the deposition mechanism a blade or a roller is common. How easy a new layer of powder is spread on the printbed is determined by the flowability of powder. This flowability describes how smooth and consistent powder moves by a force, and is influenced by particle size, particle shape and interparticle forces. If a material has smaller particles, low density and irregular shapes, it restricts the flow if you compare it to a material with a larger and uniform particle size and higher density.[23] To increase the flowability of powder, glidants can be added to modify the friction between particles of the material. Adjusting the translational velocity and rotational speed gives the powder more time for to flow and can also help to minimize defects. The better the flowability the less it affects the mechanical properties and surface quality of the tablet.

If the material has poor flowability, non-uniform distribution of the powder, and low bed packing density can occur.[12] The weight between tablets from the same print will become inconsistent, and might weigh too little. If the translational velocity is set too high, it will negatively impact the uniformity of the powder layer, resulting in a higher surface roughness and lower density of the powder.[24,25] It also introduces particle burst phenomenon, where particles break apart or deform and will lead to gaps, waviness and non-uniformities in the powder layer.[24,26]

The defects in the powder bed layer will result in bad layer adhesion between multiple layers, impacting the tablet's mechanical properties. Other issues such as tablets with varying densities may be produced from a single print batch, resulting in non-uniform weights and inconsistent API concentrations. Performing optimizations on the print parameters and the material increases the powder layer uniformity and flowability. Artificial examples of defects are shown in appendix A.

3 Materials, setup and method

3.1 Materials

The materials used in the experiments include two types of powder: mannitol 100 SD and PA12, as well as plastic filament PLA. The powders are used in the SLS printer, the PLA is used to print various models that facilitate data retrieval and are used in some of the experiments.

3.2 Experimental setup

The setup consists of two components, the powder distributor and the machine that controls it. This setup was chosen to create additional space around the powder distributor to install the 3D scanner. The powder distributor is interchangeable and originally from a SnowWhite2 SLS machine that uses a blade as a recoater. An aluminum ISB profile is installed on the side of the powder distributor. On this aluminum profile a custom module is installed and controlled by a microcontroller board. This module can tilt up and down and rotate 360 degrees. Benchmarks will indicate what movements will be used when retrieving data, and how much movement is required to make the scans as accurate as possible. This module has a mount for the 3D scanner and when installed, the 3D scanner points to the center of the printbed. A microcontroller board with an MEGA328P controls the movement of the rotation module. The Arduino IDE software is used to program and compile the code for the microcontroller. The Revopoint mini and Revo Scan 5.5.2 software are used to capture the 3D scans and generate the point clouds.

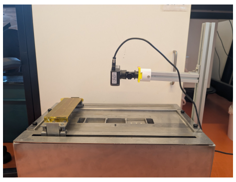

Figure 2: Setup with powder distributor, 3D scanner and rotation module

3.3 Defect detection model

To accurately analyze the point cloud, a workflow is designed to process the data consistently and automatically. For the method validation, a model will read the point cloud data produced by the 3D scanner, and process it in such a way that multiple metrics can be measured related to powder bed roughness. This workflow will validate the 3D scanner and the data it produces. The goal is to detect defects and capture the roughness of the powder layers using a 3D scanner. If the experiments show that the scanner is not accurate enough or unreliable, for example the noise that the scanner produces negatively impacts the roughness, a different approach will be proposed to continue this research. Three metrics are chosen, mean roughness, root mean square roughness and maximum height difference, an explanation will be provided. These metrics provide a good initial measurement of the roughness of the powder layer. The statistical independent t-test is used to determine if a significant difference can be found between the means of two samples. The F-test is used to compare the variance of two samples for significant differences. For both tests, the p-value needs to be smaller than 0.05 to be found significant.

3.3.1 Data acquisition

The 3D scanner should be installed on the setup, and pointed towards the middle of the printbed. The height between the scanner and the printbed is 10 centimeters. The 3D scanner is calibrated with the included calibration board that came with the 3D scanner and the calibration process can be started in the Revo Scan software. The next step is to place the 3D printed window on top of the printbed and place marker points. In the Revo Scan software a scan can be made of these markers called global marker tracking. This first scans the marker points and creates a global map. If, for some reason, a few marker points are not visible, the 3D scanner can still locate its position. As soon as the 3D scanner is installed and a global marker scan is made, the first layer can be spread on the printbed, visible in step 1 of Figure 3. The setup can be controlled from the SnowWhiteV2, so the recoater speed can be adjusted. A plastic 3D printed window is put on top of the powder layer to prevent capturing unnecessary surroundings in the 3D scan. This is printed in black and not captured by the scanner because the light gets absorbed. When this is finished, a 3D scan can be made from the powder layer, this is done by using the Revo Scan software, as shown in step 2 of Figure 3. This scan will contain a collection of points that represent the powder layer, called a point cloud. The global tracking markers are removed from the point cloud, which will then be exported to a local folder. Using the code, the point cloud is processed into a height map and histogram and the metrics are calculated. When this is finished all the figures and data are automatically exported, visible in step 3 of Figure 3.

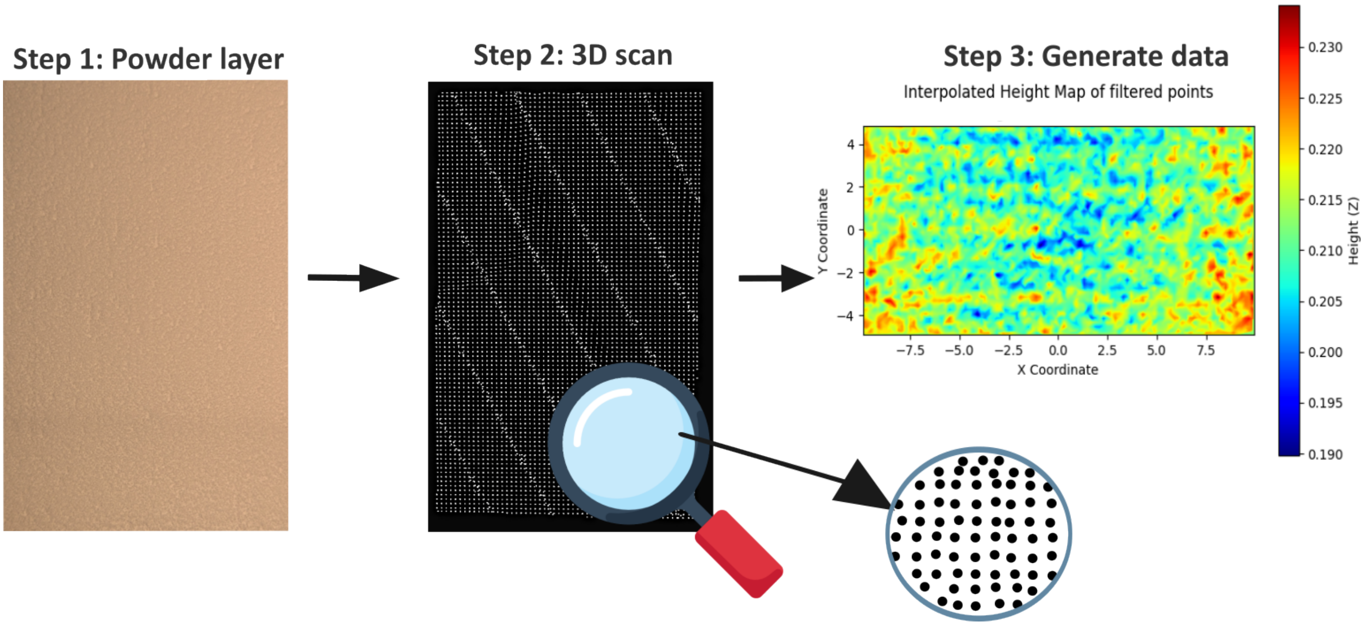

Figure 3: Workflow of workflow from capturing the powder layer to generating data on the roughness of powder layer

3.3.2 Code

A pipeline is implemented to process and analyze the 3D point clouds generated by the 3D scanner. The aim is to transform scanning data into the three surface roughness metrics and will be used to analyze the accuracy of the 3D scanner and do comparisons between different powder layer scans. Inspiration has been taken from previous research by Chen et al.[27] where point clouds were generated based on printed objects. Processing and analyzing point clouds can be computationally expensive due to the large size of the point clouds that consist of around 200,000 points. Inspiration to do power bed surface classification with little computing power was taken from previous research as well.[28] The pipeline consists of the following main operations which are executed in an automated manner: loading data, alignment with axis, filtering of data, interpolation, calculating roughness data and creating figures.

Load data and alignment: The point clouds obtained from the 3D scanner are saved into an array. Principal Component Analysis **(PCA)** is applied to make sure the X, Y and Z axes are

aligned across different point clouds.
The downside of PCA is that the main plane is used as a reference point. When a scan is made of a powder layer with large features that dominate the mean plane, or an asymmetric scan, the alignment will not work. Alternatively, PCA is applied on a local region. For the model validation, this is not an issue because the scans are made from flat surfaces without defects, but when dealing with larger features, regular PCA will not work and needs to be reconsidered, for example using the local PCA.

Filtering and outlier removal: After the alignment a region is extracted from the point cloud. This is done by calculating the middle of the point cloud and extending a window in the X and Y direction with a set value. Points outside this window are removed. With the remaining points outliers are removed and filtering is applied. This is done by using the statistical outlier filter which will remove points that deviate significantly with their neighbors. The 40 nearest neighbors are used and a standard deviation of 1.5 is used. The radius outlier removes points that have too little neighbors. A point must have at least 30 neighbors within a radius of 0.8 millimeters. Then a voxel grid filter is applied to downsample points in high-density regions, the structure is preserved, but the number of points is reduced. A voxel size of 0.2 millimeters is used. The values for the input parameters were chosen based on the height map visualization and have not yet been optimized.

Generation of figures and numerical data: The remaining points in the point cloud are transformed into multiple figures. The bounding box visualization shows the analyzed window within the original point cloud. The raw height map displays the point cloud within the window with colors representing their height and deviation. The interpolated height map provides a representation of the surface using interpolation. The height distribution histogram shows the distribution of the surface height. Three metrics are computed from the point cloud the mean roughness, root mean square roughness and maximum height difference. The code is included in Appendix H.

3.3.3 Roughness metrics

In order to quantify the surface roughness of a powder layer, three metrics are used: mean roughness, root mean square roughness and maximum height difference.

Mean roughness Mean roughness, or surface roughness **(RA)** indicates the average height deviation from the main plane and expresses the quality of the smoothness of the surface. The higher the surface roughness value, the rougher the surface. The formula to calculate the RA value is:

$$Ra = \frac{1}{N}\sum_{i=1}^{N} |z_i - \bar{z}| \tag{1}$$

z_i Height of pixel i on the surface
N Total number of pixels in the analyzed region
$\bar{z}$ Mean height of all pixels in the region

Root mean square roughness The root mean square roughness **(Rq)** also expresses the quality of the smoothness of the surface, but the deviations are squared and the square root is taken. This gives more weight to larger deviations such as peaks or gaps in the powder layer. The formula to calculate the Rq value is:

$$Rq = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (z_i - \bar{z})^2} \tag{2}$$

z_i Height of pixel i on the surface
N Total number of pixels in the analyzed region
$\bar{z}$ Mean height of all pixels in the region

Maximum height difference The maximum height difference expresses the maximum height deviation between the lowest and highest point on the powder layer. It can be used to analyze how different measurement parameters such as rotation or scanning duration impact the scan accuracy. The formula to calculate the maximum height difference is:

$$\text{Maximum height difference} = z_{\max} - z_{\min} \tag{3}$$

4 Results and discussion

Experiments have been conducted to test and validate the setup and hardware that will be used and to understand the optimal conditions to collect data. With the 3D scan, a point cloud is created and analyzed. Numerical and visual data will be generated to do a statistical analysis. For each scan, the mean roughness, root mean square roughness and the maximum height difference are calculated.

4.1 Roughness of mannitol compared to flat plastic plate

A powder layer of mannitol 100 SD is compared with a metal, glass and plastic plate to measure the noise produced by the 3D scanner. Excessive noise makes a measurement of the powder layer unrepresentative, because the noise is captured instead of the roughness. Understanding the noise is important because distinguishing a powder layer from a scan of a flat surface is necessary for the next part of the research. Collecting data with the metal plate was difficult, because metal reflects and refracts light randomly.[29] Because of this reflection the point cloud was more sparse in certain areas. The glass plate also did not work, because the light did not reflect and passed through, capturing the surface beneath. Therefore, the measurements of the glass and metal plate have been left out from this experiment. The data can be found in Appendix E.

Table 1: Average Surface Roughness Calculations for plastic plate and Mannitol

Surface	Mean Roughness in μm $\pm$SD	Root Mean Square Roughness in μm $\pm$SD	Max Height Difference in μm $\pm$SD
Mannitol	17.097 ± 0.091	21.181 ± 0.087	149.861 ± 2.829
Plastic plate	4.112 ± 0.078	5.154 ± 0.085	43.776 ± 0.715

Table 2: Statistical Test Results (t-test and F-test for plastic plate and Mannitol)

Comparison	**Metric**	**t-statistic**	**p-value**	**F-statistic**	**p-value**
Mannitol and Plastic	Mean Roughness	242.204	$1.8862 \cdot 10^{-16}$	1.348	0.390
	Root Mean Square Roughness	294.638	$1.987 \cdot 10^{-17}$	1.031	0.488
	Max Height Difference	81.293	$2.592 \cdot 10^{-8}$	15.674	0.010

Table 1 shows that the mean roughness, root mean square roughness and maximum height difference are lower for the plastic plate compared to mannitol. In Table 2 a significant difference between the mean values of mannitol and plastic plate is shown. This shows that little noise is produced by the 3D scanner, and does not dominate the roughness of the powder layer in the measurement. In Figure 4 the height map of mannitol and the plastic plate is shown, the difference in roughness clearly is visible.

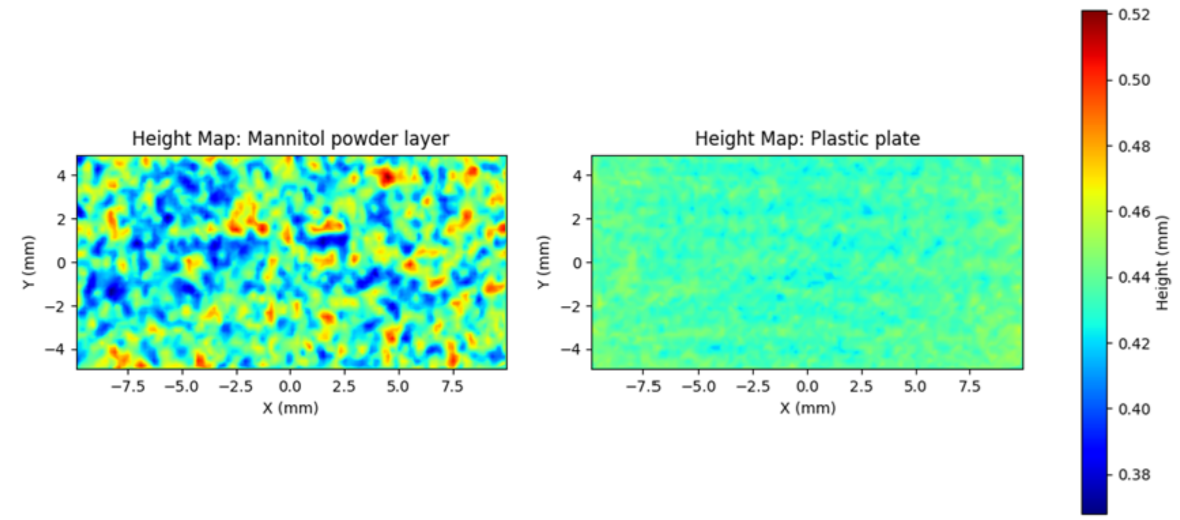

Figure 4: Height map of mannitol and plastic plate

The Shapiro-Wilk test is used to statistically test if the mannitol or plastic plate follow a normal distribution. If a p-value smaller than 0.05 is found, the null hypothesis cannot be rejected, indicating that the data is not significantly different from a normal distribution. If the p-value is found to be greater than 0.05, the null hypothesis is rejected, indicating the data does not follow a normal distribution. The histograms of the plastic plate and mannitol are shown in Figure 5 including a normal fit. Using this test, the plastic plate gave a p-value of 0.1159, the null hypothesis cannot be rejected and suggests that the data follows a normal distribution. This indicates that the deviations in the height measurements for the plastic plate data are random. Mannitol gave a p-value of 0.0001. Therefore, the null is rejected, meaning that the data does not follow a normal distribution. The height deviations are not only caused by noise from the 3D scanner, but also by powder properties and deposition of the powder. This shows that the roughness that is captured by the 3D scanner is influenced by the powder properties, rather than being caused only by noise.

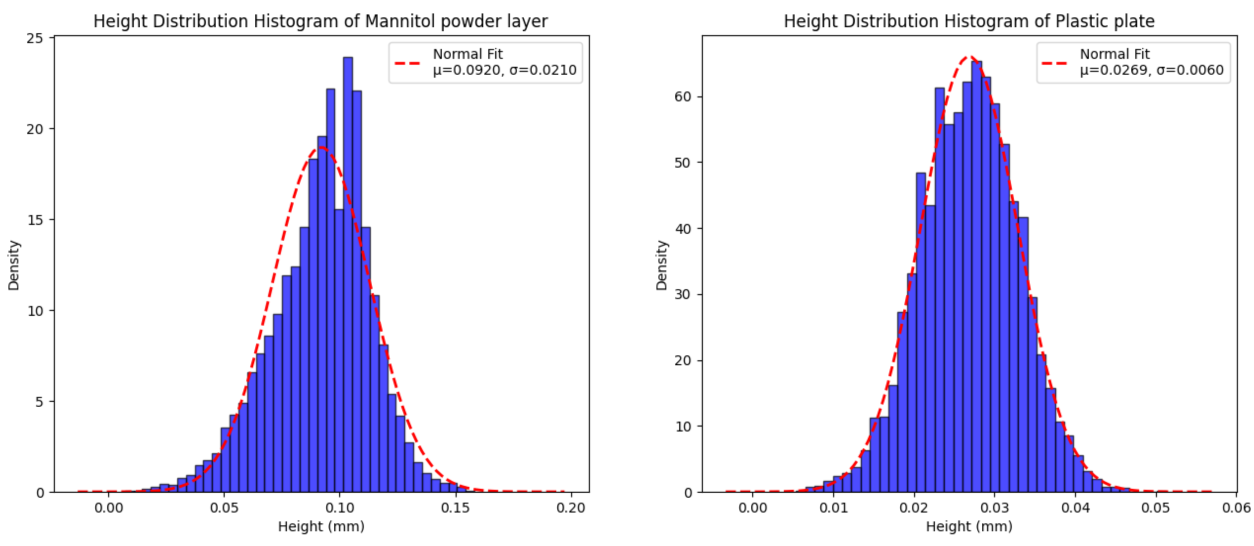

Figure 5: Height histogram of Mannitol powder layer and plastic plate

4.2 Stationary and rotational scan accuracy

This experiment is done to understand the effect of rotating the scanner or having the scanner remain stationary during scanning. The hypothesis is that rotation during scanning provides more information about the Z-coordinate, reducing the maximum height difference and resulting in a more accurate measurement. This experiment includes a stationary scan, a scan while rotating the scanner 10 degrees using the rotation module, and manually rotating the scanner. All measurements were taken on a flat powder layer of mannitol, so the deviations in the measurements can be assumed to be noise, and therefore a comparison can be made between the stationary and rotational scans. The data can be found in Appendix D.

Table 3: Average Surface Roughness Values for stationary and rotational 3D scans

Class	Mean Roughness in μm $\pm$SD	Root Mean Square Roughness in μm $\pm$SD	Max Height Difference in μm $\pm$SD
Stationary	15.118 ± 0.051	19.210 ± 0.066	170.771 ± 1.126
Module rotation	13.811 ± 0.416	17.563 ± 0.573	170.049 ± 50.970
Manual rotation	13.435 ± 0.577	17.055 ± 0.728	143.710 ± 4.699

As shown in Table 4, a significant difference was found between stationary and manual rotation, with the manual rotation scans being more accurate. This indicates that more information is gathered while rotating the 3D scanner, resulting in more accurate scans. The mean values were also all found to be lower for the manual rotation compared to the stationary measurements, visible in Table 3. The difference between the mean max height was the largest, indicating that the height can be measured more accurately because the angled view provided more information about the surface variations. Between the stationary and rotation module, a statistical difference was found for the mean roughness and the root mean roughness, the maximum height difference was not found to be significant. Although rotating the scanner during measurement improved the accuracy of the scans, it also introduced inconsistencies in the data collection. The rotation module was sensitive and slightly moved during scanning, resulting in unusable measurements. This explains the high standard deviation for the max height difference for the module rotation, shown in Table 3. If the rotation module remains part of the setup, it must be made more robust to make it consistent during measurements. Although a significant difference was found between stationary and the module and manual rotation, it also remains unknown what processing is done by the 3D scanner during rotation. It is possible that the scanner performs additional processing during rotation, increasing the accuracy of the scans rather than the rotation itself.

Table 4: Statistical Test Results (t-test and F-test for Stationary and 2 rotation types)

Comparison	**Metric**	**t-statistic**	**p-value**	**F-statistic**	**p-value**
Stationary and Rotation module	Mean Roughness	6.968	$1.9966 \cdot 10^{-3}$	0.015	0.999
	Root Mean Square Roughness	6.383	0.003	0.013	0.999
	Max Height Difference	0.0323	0.976	0	1
Stationary and Manual rotation	Mean Roughness	6.498	$2.7404 \cdot 10^{-3}$	0.008	0.9998
	Root Mean Square Roughness	6.591	0.003	0.008	0.9998
	Max Height Difference	12.522	$1.213 \cdot 10^{-4}$	0.057	0.991
Module and Manual rotation	Mean Roughness	1.181	0.275	0.521	0.728
	Root Mean Square Roughness	1.226	0.257	0.620	0.673
	Max Height Difference	1.151	0.313	117.645	$2.119 \cdot 10^{-4}$

4.3 Impact of environmental light on accuracy

This experiment was performed to determine the impact of making 3D scans with or without environmental light. The hypothesis is that this impacts the accuracy of the scan by light pollution. However, it is unknown how much this impacts the noise and quality of the 3D scan. A total of 12 measurements were taken in the same room. To simulate the effect of environmental light, six scans were taken with the lights off, and six with the lights on. The used material was Mannitol. The data can be found in Appendix B.

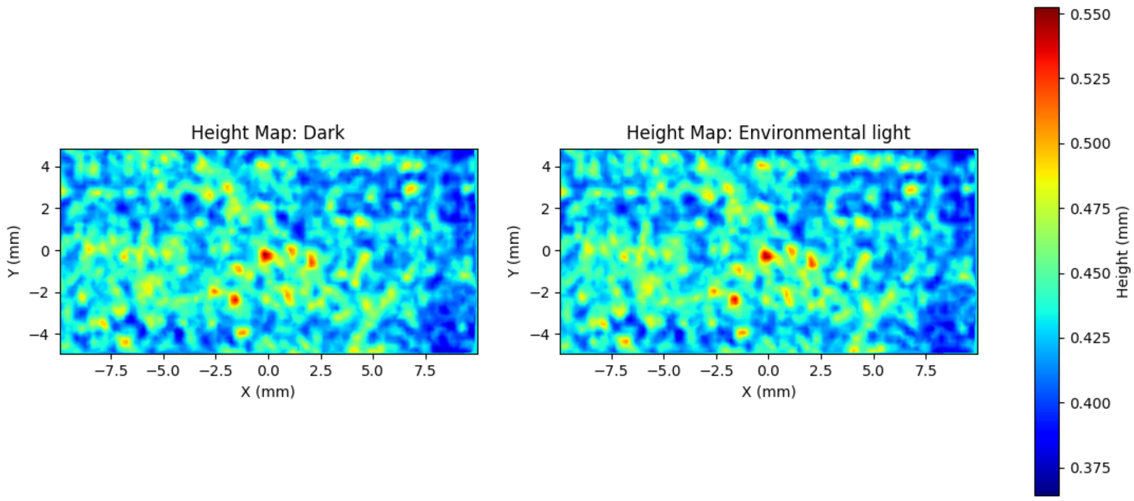

Figure 6: Height map of scan in room with lights off and on

No visual differences can be observed in the height map shown in Figure 6. In Table 5, all p-values were greater than 0.05, so there is no significant difference in the mean value for the 3 measures metrics. For the variance, no statistical difference was found as well, indicating that neither conditions are more stable around the mean or the variance. This shows that environmental light does not significantly impact the accuracy of the 3D scans, and does not have to be controlled during the measurements.

Table 5: Statistical Test Results (t-test and F-test for impact environmental light)

Metric	**t-statistic**	**p-value**	**F-statistic**	**p-value**
Mean Roughness	-0.069	0.947	0.9640	0.516
Root Mean Square Roughness	-0.073	0.943	0.979	0.501
Max Height Difference	0.197	0.848	1.757	0.276

4.4 Impact of scan time duration on accuracy

To measure the effect of the scan duration on the results, another experiment was conducted. The hypothesis is that the accuracy of the scans will improve because the scanner has more time to process and correct outliers and noise. A single powder layer was spread on the printbed and three different scan times were performed, 5 seconds, 10 seconds and 20 seconds scan time. For every class a total of five measurements were taken. The data can be found in Appendix C.

Table 6: Average Surface Roughness Values for Different Scan Times

Scan Time	Mean Roughness in μm $\pm$SD	Root Mean Square Roughness in μm $\pm$SD	Max Height Difference in μm $\pm$SD
5 seconds	19.145 ± 0.02	24.027 ± 0.022	188.701 ± 3.293
10 seconds	19.097 ± 0.053	23.931 ± 0.064	179.111 ± 2.481
20 seconds	18.975 ± 0.019	23.783 ± 0.026	177.737 ± 2.152

In Figure 7, the difference in height map for 5 second and 20 second scan time are shown. Similar to the results in Table 6 the mean roughness and root mean square roughness show minimal differences. However, the highest and lowest points in the scan are closer to the mean value in the 20 seconds scan. This shows that the scanner has more time to process and correct outliers and noise, which are smoothed out when the scan time is longer.

Table 7: Statistical Test Results (T-tests for Different Scan Times)

Comparison	**Metric**	**t-statistic**	**p-value**	**F-statistic**	**p-value**
5 and 10 seconds	Mean Roughness	1.753	0.129	0.292	0.870
	Root Mean Square Roughness	3.192	0.025	0.120	0.978
	Max Height Difference	5.202	0.001	1.762	0.298
10 and 20 seconds	Mean Roughness	4.833	$4.677 \cdot 10^{-3}$	7.662	0.037
	Root Mean Square Roughness	4.786	$4.325 \cdot 10^{-3}$	6.175	0.053
	Max Height Difference	0.936	0.038	1.329	0.395
5 and 20 seconds	Mean Roughness	10.9668	$1.1884 \cdot 10^{-5}$	2.241	0.227
	Root Mean Square Roughness	16.066	$2.8320 \cdot 10^{-7}$	0.744	0.606
	Max Height Difference	6.2327	$4.5944 \cdot 10^{-4}$	2.341	0.215

In Table 7, a significant difference can be seen between scan duration of 10 compared to 20 seconds, as well as between 5 and 20 seconds for all three metrics. This indicates that scan time influences the accuracy of the scan, reducing the noise and outliers. Two of the three means between 5 and 10 seconds were found to be significant, but the difference between the means shown in Table 6 is found to be minimal. This experiment has shown the impact of the scan time duration. While a significant difference was found between a scan duration of 10 seconds and 20 seconds, the difference in mean values is minimal. For future measurements, a scan duration of 10 seconds will be used, providing a good balance between accuracy and time efficiency.

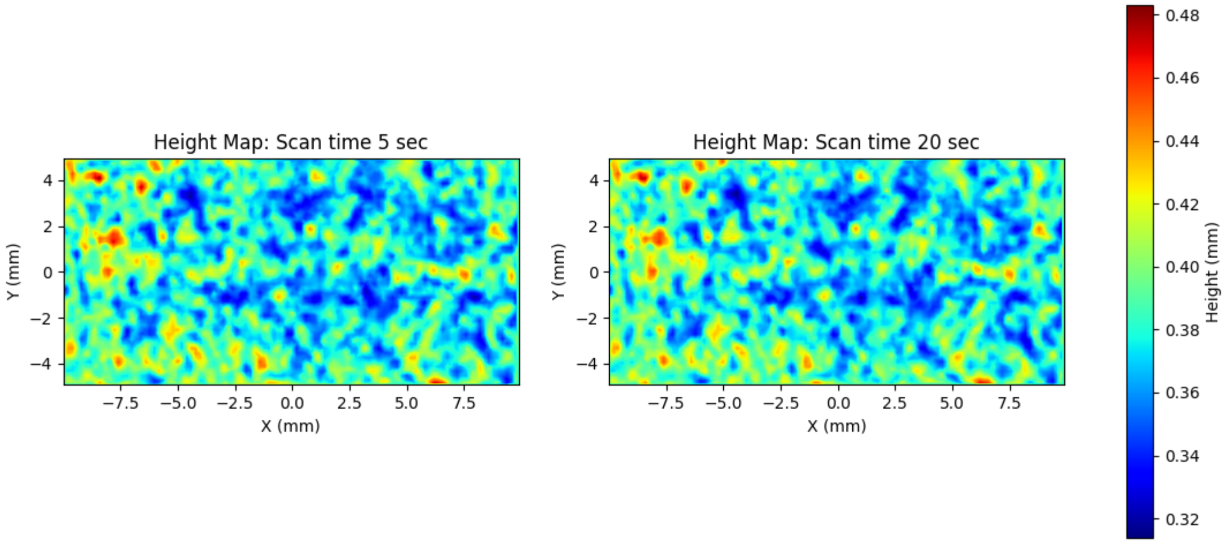

Figure 7: Height map of 5 seconds and 20 seconds scan time

4.5 Roughness of powders

This experiment is performed to show that the difference in roughness of two different powders can be captured by the 3D scanner. The materials used are Nylon (PA12) and sugar alcohol (Mannitol). PA12 has smaller particles compared to mannitol, and is therefore expected to spread smoother powder layers, as it has a smaller particle size. A total of ten measurements were taken, five for each material. The data can be found in Appendix F.

Table 8: Average surface roughness calculations for Mannitol, PA12 and plastic plate

Surface	Mean Roughness in μm ±SD	Root Mean Square Roughness in μm ±SD	Max Height Difference in μm ±SD
Mannitol powder layer	13.797 ± 0.125	17.815 ± 0.138	167.224 ± 3.751
PA12 powder layer	8.105 ± 0.027	10.091 ± 0.030	69.470 ± 2.180
Plastic plate	4.112 ± 0.078	5.154 ± 0.085	43.776 ± 0.715

Additionally, the mean values for the plastic plate are included in Table 8 from the experiment in Section 4.1. The data from the plastic playte shows that the roughness of both powders is much higher than the noise on the plastic plate. As seen in Table 9, a significant difference is found between PA12 and Mannitol for each metric. The average roughness, root mean square roughness and maximum height difference are lower for the PA12 in comparison to Mannitol, as shown in Table 8. This indicates that it is possible to measure the roughness of different materials with the 3D scanner and distinguish powders based on their roughness. Figure 8 shows a visual difference as well, where the PA12 powder has a more uniform layer compared to Mannitol.

Table 9: Statistical test results (t-test and F-test for Mannitol and PA12 powder layer)

Metric	**t-statistic**	**p-value**	**F-statistic**	**p-value**
Mean Roughness	99.754	$1.6577 \cdot 10^{-8}$	21.524	$5.738 \cdot 10^{-3}$
Root Mean Square Roughness	122.167	$6.998 \cdot 10^{-9}$	21.899	$5.554 \cdot 10^{-3}$
Max Height Difference	50.38	$1.365 \cdot 10^{-9}$	2.960	0.159

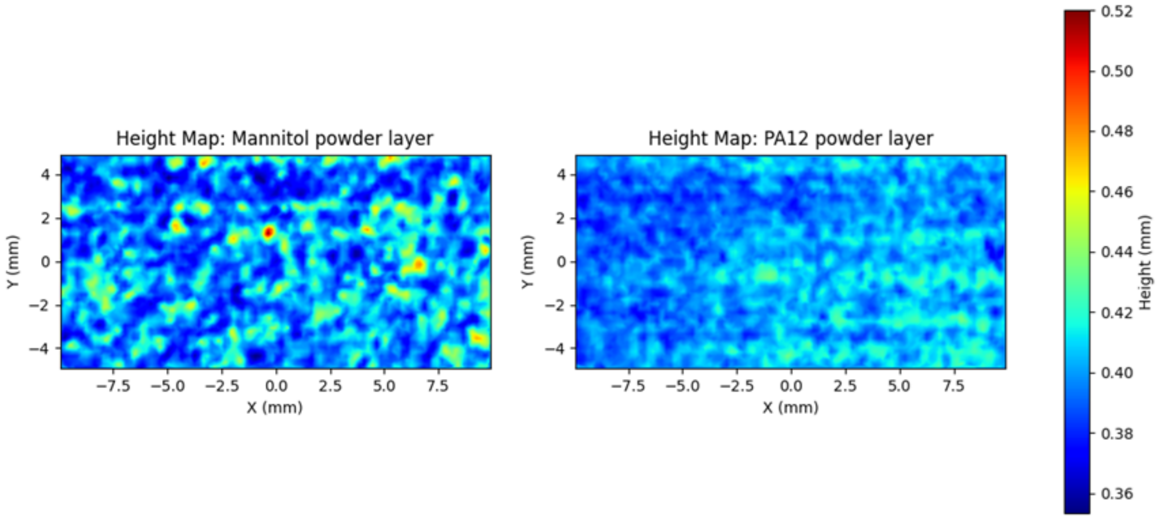

Figure 8: Height map of Mannitol and PA12 powder layer

4.6 Measured height difference between 2 consecutive steps

A 3D printed staircase was printed to verify if the 3D scanner can accurately measure the difference in height between every step. The difference in height between the two steps was designed to be 0.3 millimeters. The height map and histogram from this scan are shown in Figure 9. In the histogram eight distinct peaks are visible, each representing one step. The difference in height in the height map is also clearly visible.

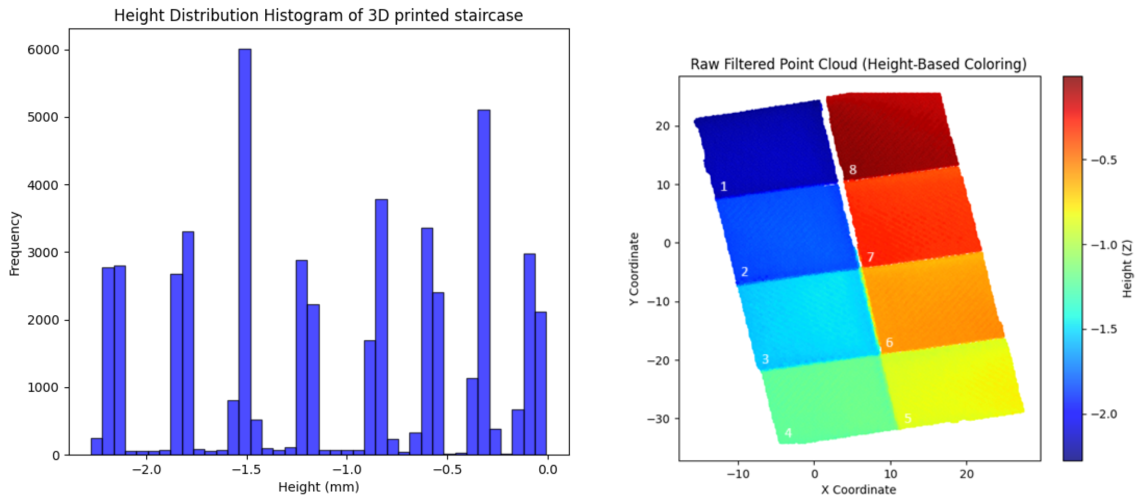

Figure 9: Height histogram and height map of 3D printed staircase

To understand how accurately the 3D scanner can measure height differences between each step, the difference in step size was measured in both the 3D primted object and 3D scan. The results are in Table 10. The difference between the two steps in the 3D printed object was close to the set 0.3 millimeters, with some small deviations. The 3D scan data shows that the measured height difference are smaller than 0.3 millimeters for the steps one to four, and slightly larger than 0.3 millimeters for the steps five to eight. This indicates a potential defect or artifact that happened during scanning, suggesting that the 3D scanner was slightly tilted towards one side of the printbed. Despite this, the variations were small and all the values were within 0.05 millimeters of the true height difference, which is within the minimum resolution of the 3D scanner.

Table 10: Difference in height between two steps

Height difference between 2 steps	3D object in mm	3D scan in mm
Step 1 and 2	0.3	0.25
Step 2 and 3	0.29	0.25
Step 3 and 4	0.3	0.28
Step 4 and 5	0.3	0.34
Step 5 and 6	0.27	0.31
Step 6 and 7	0.3	0.32
Step 7 and 8	0.29	0.33

4.7 Conclusion

The experiments gave more insight into the accuracy and noise produced by the 3D scanner and the optimal scanning conditions. Scanning the plastic plate and mannitol gave insight on the produced noise by the 3D scanner. It was shown that the scanner is accurate enough to distinguish between different powders. The noise generated by the 3D scanner did not limit the accuracy of the measurements. The experiment where two different powders were compared with different properties showed that it is possible to distinguish the roughness of powders based on the data. The remaining experiments provided more information about the conditions to collect the most accurate data. Environmental light does not have an impact. Rotating the 3D scanner during scanning does impact the accuracy of the data, manual rotation showed the biggest improvement in rotation. Manual rotation is inconsistent and the rotation module needs to be made more robust. If no solution can be found, rotation will not be used in the future even though the rotation improved the result, the data is already good enough to make the distinction between two different powders. The scan time improved the accuracy of the scan as well, while the scans with duration of 20 seconds were the most accurate, the difference between 10 and 20 seconds was minor, so for future measurements, a scan duration of 10 seconds will be used to make data gathering faster. The 3D scanner was able to measure the height difference with minimal deviations. Artifacts in the data showed that the scanner was slightly tilted, the variations were within the resolution of the scanner. Future experiments will be conducted in a room with the lights on for convenience, using a scan time of 10 seconds, without rotating the 3D scanner.

5 Outline of Part 2

The optimal scan conditions are found during the model validation to make accurate measurements, and the designed workflow shows that it is possible to convert powder layers into point cloud data that can and will be used in Part 2. A pipeline will be developed that can perform feature recognition in the data. Examples of the features are visible in Appendix A. This pipeline will consist of a neural network and first needs be designed and trained. An option would be to use DenseNet from the PyTorch package in python. To train and evaluate the neural network, the dataset needs to be labeled first. The neural network also needs to be designed for the point cloud data and the feature extraction task. The performance of the network will be measured using accuracy, precision, recall, F1-score. To validate the network, the labeled dataset will be split into a train and validation set, k-fold cross-validation could be a good option.

5.1 Research question(s)

Main research question:

Can neural networks be used to classify defects in powder layer?

Subquestions Part 2:

- What kind of pre-processing is necessary to convert raw point cloud data into a usable format for the input of the neural network?
- What kind of neural network architecture fits the best for processing and analyzing point cloud data?
- How to label the features in point cloud data efficiently?
- How accurate can a neural network recognize and classify defects in point cloud data?

6 Ethics and Privacy Quick Scan

"The Ethics and Privacy Quick Scan of the Utrecht University Research Institute of Information and Computing Sciences was conducted (see Appendix G)." No issues found: "It classified this research as low-risk with no fuller ethics review or privacy assessment required."

7 Suggestion for camera

Previous literature has used cameras and 3D scanners to test and measure the powder layer quality and spreadability of powder. This research will conduct ex-situ spreadability testing using a 3D scanner, but in case the 3D scanner is ineffective, a camera will be used as an alternative to continue performing the analysis of the powder layer. An overview of existing literature will be provided about cameras or 3D scanners to do powder spreadability analysis. Based on this overview, a recommendation will be made for the camera to purchase. Table 11 contains an overview of the type of hardware that is used and the goal of the research.

Table 11: Overview powder spreadability testing in literature

Type of hardware	Goal
Contact image sensor from Canon LiDE 220 flatbed scanner[19]	Measure layer thickness, surface roughness and particle density
Tripod-mounted Canon 700D camera (1855mm lend, automatic mode)[30]	Coverage area of powder using blade
Point Grey GS3-U3-50S5M-C[31]	Quantification of powder bed
Nikon EOS Rebel T6i[32]	Surface coverage, Surface roughness
Thorlabs CMOS Camera DCC3240M, Thorlabs Coaxial, Zoom Lens MVL6X3Z, Extension Tube MVL05A, C-Mount Adapter MVL-CMC[33]	Layer uniformity
Mikrotron 1362 camera[34]	Surface roughness
Revopoint mini 3D scanner and CCD camera (GRAS-20S4M-C, 2.0 MP, 30 FPS, monochromatic) with lens(LM35JC, KOWA)[35]	Powder bed density, surface uniformity and spreading forces

A camera allows the retrieval of qualitative data[35] of the powder bed and is used for visual inspections of defects, powder spreadability and uniformity of the powder bed. Using a camera has downsides such as a limited spatial resolution or small field-of-view **(FOV)**.[19] It makes it more difficult to detect layer thickness variations on the bed and powder packing density. Using a contact image sensor solves the small FOV and low-resolution problems, but still has issues with blurriness, focussing and depths.

A 3D scanner allows the retrieval of quantitative data[35] of the powder bed. It can be used to create a point cloud. Point clouds enable to possibility to build height maps and gather data about the height deviation of each point in the cloud. This can be used to calculate the uniformity of the powder bed, local layer thickness and roughness of the powder layer **(RA)**. A camera or 3D scanner allows different analyses on the powder bed layer.[35]

Recommendation: This research focuses on powder layer defect detection and flowability of powder, based on Table 11 previous research that is quite similar are.[30,32] Because of this, one of the newer models of the Nikon EOS Rebel T6i or Canon 700D are recommended to purchase. The newer Canon EOS R50 or Canon EOS R100 would be recommended. The R100 with lens is cheaper compared to the R50 that comes with the body only.

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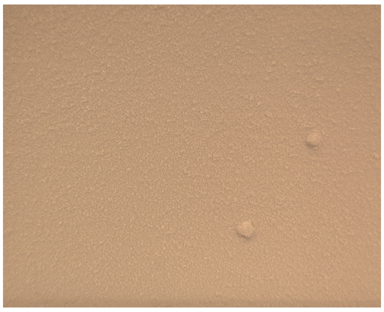
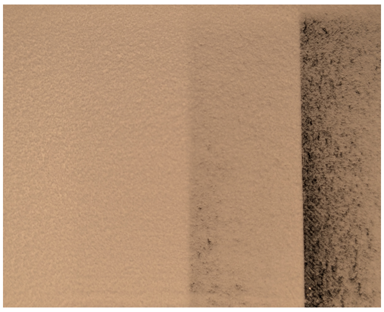
Appendix A Artificial examples of defects

(a) Rough surface

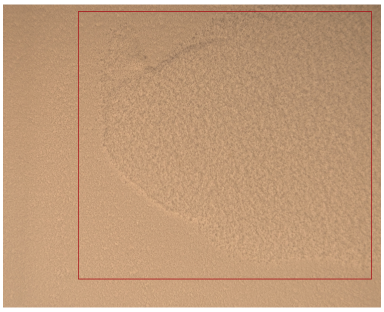
(b) Cracks



(c) Holes



(d) Non-uniform extreme case



(e) Non-uniform

(f) Minor crack bottom

Figure 10: Artificial examples of defects in the powder bed layer

Appendix B Experiment data: light and dark room

Table 12: Surface Roughness Calculations

Class	Filename	Mean Roughness in μm	Root Mean Square Roughness in μm	Max Height Difference in μm
Dark	01_pc.ply	19.337	24.716	197.396
Dark	02_pc.ply	19.336	24.679	189.396
Light	03_pc.ply	19.409	24.789	189.377
Light	04_pc.ply	19.465	24.855	191.582
Light	05_pc.ply	17.551	22.046	175.543
Light	06_pc.ply	17.460	22.013	171.473
Dark	07_pc.ply	17.454	22.010	167.938
Dark	08_pc.ply	17.445	21.951	172.287
Light	09_pc.ply	15.832	20.334	202.632
Light	10_pc.ply	15.754	20.214	203.877
Dark	11_pc.ply	15.771	20.217	210.522
Dark	12_pc.ply	15.743	20.161	207.683

Appendix C Experiment Data: Different Scan Times

Table 13: Surface Roughness Calculations for Different Scan Times

Class	Filename	Mean Roughness in μm	Root Mean Square Roughness in μm	Max Height Difference in μm
5s	13.ply	19.177	24.049	190.423
5s	14.ply	19.168	24.052	184.601
5s	15.ply	19.146	24.023	188.899
5s	16.ply	19.110	24.004	186.538
5s	17.ply	19.122	24.009	193.045
10s	18.ply	19.187	24.036	176.341
10s	19.ply	19.064	23.883	179.164
10s	20.ply	19.067	23.886	178.267
10s	21.ply	19.105	23.946	183.115
10s	22.ply	19.063	23.902	178.669
20s	23.ply	18.944	23.741	176.577
20s	24.ply	18.992	23.805	180.998
20s	25.ply	18.990	23.802	178.551
20s	26.ply	18.974	23.778	175.375
20s	27.ply	18.975	23.788	177.185

Appendix D Experiment Data: Stationary and rotational

Table 14: Surface Roughness Calculations for stationary and rotational 3D scans

Class	Filename	Mean Roughness in μm	Root Mean Square Roughness in μm	Max Height Difference in μm
Stationary	32.ply	15.119	19.242	172.211
Stationary	33.ply	15.031	19.094	171.292
Stationary	34.ply	15.135	19.227	170.370
Stationary	35.ply	15.162	19.258	170.810
Stationary	36.ply	15.143	19.229	169.170
Rotation module	37.ply	13.660	17.397	148.940
Rotation module	38.ply	13.952	17.709	149.248
Rotation module	39.ply	13.413	17.031	149.216
Rotation module	40.ply	14.467	18.483	261.050
Rotation module	41.ply	13.562	17.194	141.793
Rotation manual	42.ply	12.728	16.126	137.825
Rotation manual	43.ply	14.192	17.998	150.620
Rotation manual	44.ply	13.375	17.103	143.839
Rotation manual	45.ply	13.797	17.450	144.756
Rotation manual	46.ply	13.084	16.596	141.510

Appendix E Experiment Data: Metal plate, plastic plate and mannitol powder layer

Table 15: Surface Roughness Calculations for metal, plastic and powder surface

Class	Filename	Mean Roughness in μm	Root Mean Square Roughness in μm	Max Height Difference in μm
Powder layer mannitol	62.ply	17.234	21.289	153.107
Powder layer mannitol	63.ply	17.078	21.198	148.505
Powder layer mannitol	64.ply	17.135	21.227	147.128
Powder layer mannitol	65.ply	17.034	21.117	147.853
Powder layer mannitol	66.ply	17.005	21.072	152.714
Flat metal plate	67.ply	10.825	13.488	156.072
Flat metal plate	68.ply	11.076	13.764	160.971
Flat metal plate	69.ply	11.218	14.039	146.442
Flat metal plate	70.ply	11.178	14.021	140.475
Flat metal plate	71.ply	11.454	14.422	144.587
Flat plastic plate	72.ply	4.206	5.258	44.053
Flat plastic plate	73.ply	4.180	5.221	44.308
Flat plastic plate	74.ply	4.060	5.110	43.595
Flat plastic plate	75.ply	4.088	5.133	44.315
Flat plastic plate	76.ply	4.024	5.047	42.610
Glass plate	Laser went through glass plate	-	-	-

Appendix F Experiment Data: Mannitol and PA12 powder layer

Table 16: Surface Roughness Calculations for mannitol and PA12 powder layer

Class	Filename	Mean Roughness in μm	Root Mean Square Roughness in μm	Max Height Difference in μm
Mannitol powder later	82.ply	13.929	17.936	166.552
Mannitol powder later	83.ply	13.902	17.961	167.362
Mannitol powder later	84.ply	13.773	17.773	161.779
Mannitol powder later	85.ply	13.618	17.621	168.176
Mannitol powder later	86.ply	13.763	17.782	172.249
PA12 powder layer	87.ply	8.139	10.139	67.415
PA12 powder layer	88.ply	8.088	10.084	72.453
PA12 powder layer	89.ply	8.071	10.058	68.265
PA12 powder layer	90.ply	8.106	10.084	71.093
PA12 powder layer	91.ply	8.122	10.091	68.126

Appendix G Ethics review

Response Summary:

Section 1. Research projects involving human participants

P1. Does your project involve human participants? This includes for example use of observation, (online) surveys, interviews, tests, focus groups, and workshops where human participants provide information or data to inform the research. If you are only using existing data sets or publicly available data (e.g. from X, Reddit) without directly recruiting participants, please answer no.

- No

Section 2. Data protection, handling, and storage

The General Data Protection Regulation imposes several obligations for the use of **personal data** (defined as any information relating to an identified or identifiable living person) or including the use of personal data in research.

D1. Are you gathering or using personal data (defined as any information relating to an identified or identifiable living person)?

- No

Section 3. Research that may cause harm

Research may cause harm to participants, researchers, the university, or society. This includes when technology has dual-use, and you investigate an innocent use, but your results could be used by others in a harmful way. If you are unsure regarding possible harm to the university or society, please discuss your concerns with the Research Support Office.

H1. Does your project give rise to a realistic risk to the national security of any country?

- No

H2. Does your project give rise to a realistic risk of aiding human rights abuses in any country?

- No

H3. Does your project (and its data) give rise to a realistic risk of damaging the University's reputation? (E.g., bad press coverage, public protest.)

- No

H4. Does your project (and in particular its data) give rise to an increased risk of attack (cyber- or otherwise) against the University? (E.g., from pressure groups.)

- No

H5. Is the data likely to contain material that is indecent, offensive, defamatory, threatening, discriminatory, or extremist?

- No

H6. Does your project give rise to a realistic risk of harm to the researchers?

- No

H7. Is there a realistic risk of any participant experiencing physical or psychological harm or discomfort?

- No

H8. Is there a realistic risk of any participant experiencing a detriment to their interests as a result of participation?

- No

H9. Is there a realistic risk of other types of negative externalities?

- No

Section 4. Conflicts of interest

C1. Is there any potential conflict of interest (e.g. between research funder and researchers or participants and researchers) that may potentially affect the research outcome or the dissemination of research findings?

- No

C2. Is there a direct hierarchical relationship between researchers and participants?

- No

Section 5. Your information.

This last section collects data about you and your project so that we can register that you completed the Ethics and Privacy Quick Scan, sent you (and your supervisor/course coordinator) a summary of what you filled out, and follow up where a fuller ethics review and/or privacy assessment is needed. For details of our legal basis for using personal data and the rights you have over your data please see the University's privacy information. Please see the guidance on the ICS Ethics and Privacy website on what happens on submission.

Z0. Which is your main department?

- Information and Computing Science

Z1. Your full name:

Lukas Vaartjes

Z2. Your email address:

l.vaartjes@students.uu.nl

Z3. In what context will you conduct this research?

- As a student for my master thesis, supervised by::
  Project: Sanne Abeln, Daily: Wessel Kooijman

Z5. Master programme for which you are doing the thesis

- Computing Science

Z6. Email of the course coordinator or supervisor (so that we can inform them that you filled this out and provide them with a summary):

s.abeln@uu.nl

Z7. Email of the moderator (as provided by the coordinator of your thesis project):

coordinator.cosc@uu.nl

Z8. Title of the research project/study for which you filled out this Quick Scan:

Image analysis of powder surfaces created with powder coaters using point cloud generated by 3D scanner / images by camera

Z9. Summary of what you intend to investigate and how you will investigate this (200 words max):

Drugs are ineffective for 50% of the patients when produced in mass production.]3D printing, in this case powder bed fusion, can be used to allow personalized drug production. The issue with personalized drug production is the fact that not all materials are suitable for the production of tablets. For example, selective laser sintering (SLS) allows the production of customized parts and is used in industries such as aerospace and automotive. The material used for SLS printing must have specific requirements such as particle size, shape, and flow properties to ensure that the print process will be successful. In pharmaceuticals, the materials used to produce drugs are generally not suitable for SLS manufacturing. Problems such as uneven dosage distribution or weak layer adhesion in the tablet can occur. Pharmaceutical materials often require modification to have successful result while using SLS.

The expected outcome of this research is a tool to do an analysis of a layer of powder and provide more information about this layer. Further investigation is required to determine when a yes or a no is provided and why. Data will be gathered with a camera and/or 3D scanner, and analysis will be done on this data. Specific implementation will be determined during part 1, but examples would be indicating the surface roughness to the user, or show where problems occurred in the powder layer.

Z10. In case you encountered warnings in the survey, does supervisor already have ethical approval for a research line that fully covers your project?

- Not applicable

Scoring

- Privacy: 0
- Ethics: 0

Appendix H Code

```
import os
import numpy as np
import matplotlib.pyplot as plt
from pyntcloud import PyntCloud
from sklearn.decomposition import PCA
from scipy.interpolate import griddata
import open3d as o3d
from scipy.stats import norm, shapiro


# Load point cloud and apply PCA
def load_and_align_point_cloud(folder, name_data):
    print(f"Loading point cloud data from: {folder}/{name_data}")
    pointcloud = PyntCloud.from_file(f"./code/data/{folder}/{name_data}")
    df = pointcloud.points
    X, Y, Z = df["x"].values, df["y"].values, df["z"].values
    print("Load points from point cloud")

    # Apply PCA
    points = np.column_stack((X, Y, Z))
    pca = PCA(n_components=3)
    pca.fit(points)
    aligned_points = pca.transform(points)

    # Flip transformed points around the horizontal axis and negate Z-values
    aligned_points[:, 1] = -aligned_points[:, 1]
    aligned_points[:, 2] = -aligned_points[:, 2]
    print("Aligned and transformed point cloud with PCA")

    return aligned_points

# Filter points by determining center and expand horizontally and vertically
def filter_points_by_pca_window(aligned_points, center_x, center_y, window_x,
                                window_y):
    pca_x_min, pca_x_max = center_x - window_x, center_x + window_x
    pca_y_min, pca_y_max = center_y - window_y, center_y + window_y
    print(f"Filtering points within the window: X = [{pca_x_min}, {pca_x_max}
                                          ], Y = [{pca_y_min}, {
                                          pca_y_max}]")

    filtered_mask = (
        (aligned_points[:, 0] > pca_x_min) & (aligned_points[:, 0] <
                                              pca_x_max) &
        (aligned_points[:, 1] > pca_y_min) & (aligned_points[:, 1] <
                                              pca_y_max)
    )

    filtered_pca_xyz = aligned_points[filtered_mask]
    print(f"Points after border filtering: {len(filtered_pca_xyz)}")

    # Shift all Z-values so the lowest Z-value becomes 0, Ensure there are
                                          points in the bounding box
    if len(filtered_pca_xyz) > 0:
        z_min_inside_bbox = np.min(filtered_pca_xyz[:, 2])
        # Shift Z-values
        filtered_pca_xyz[:, 2] -= z_min_inside_bbox

    return filtered_pca_xyz, filtered_mask, pca_x_min, pca_x_max, pca_y_min,
                                          pca_y_max

# Filter in point cloud by using downsampling
def filter_point_cloud(filtered_pca_xyz, point_cloud_filtering):

    if point_cloud_filtering is False:
        return filtered_pca_xyz
```

```
    # Convert to Open3D point cloud
    filtered_pcd = o3d.geometry.PointCloud()
    filtered_pcd.points = o3d.utility.Vector3dVector(filtered_pca_xyz)

    # Statistical outlier removal (remove if significant deviate from
                                          neihgbours)
    if point_cloud_filtering and len(filtered_pca_xyz) > 0:
        cl, ind = filtered_pcd.remove_statistical_outlier(nb_neighbors=40,
                                              std_ratio=1.5)
        filtered_pcd_stat_out = filtered_pcd.select_by_index(ind)
        print(f"Points after statistical outlier removal: {len(ind)}")
    else:
        print("Skipping statistical outlier removal, 0 points")
        filtered_pcd_stat_out = filtered_pcd

    # Radius outlier removal (remove if nog enough neighbours)
    if point_cloud_filtering and len(filtered_pcd_stat_out.points) > 0:
        cl, ind = filtered_pcd_stat_out.remove_radius_outlier(nb_points=30,
                                              radius=0.8)
        filtered_pcd_radius_out = filtered_pcd_stat_out.select_by_index(ind)
        print(f"Points after radius outlier removal: {len(ind)}")
    else:
        print("Skipping radius outlier removal, 0 points")
        filtered_pcd_radius_out = filtered_pcd_stat_out

    # Voxel Downsampling
    voxel_size = 0.2  # More points will remained with inctreased value
    if point_cloud_filtering and len(filtered_pcd_radius_out.points) > 0:
        downsampled_pcd = filtered_pcd_radius_out.voxel_down_sample(
                                              voxel_size)
        print(f"Points after voxel downsampling: {len(downsampled_pcd.points)
                                              }")
    else:
        print("Skipping voxel downsampling, 0 points")
        downsampled_pcd = filtered_pcd_radius_out

    return np.asarray(downsampled_pcd.points)

# Create and save figures
def save_height_map_figures(X_filtered, Y_filtered, Z_filtered,
                                          grid_z_flipped, output_dir, Ra,
                                          aligned_points, filtered_mask,
                                          pca_x_min, pca_x_max, pca_y_min,
                                          pca_y_max):

    # Figure 1 Point Cloud with Height Map Overlay
    print("Saving Figure 1: Original Point Cloud with bounding box on window"
                                          )
    fig, ax = plt.subplots(figsize=(7, 6))
    # Plot all original points in light gray
    ax.scatter(aligned_points[:, 0], aligned_points[:, 1], c='lightgray', s=1
                                          , alpha=0.5, label="Original
                                          Points (Outside BBox)")
    # Select only the points inside the bounding box
    inside_bbox_mask = (
        (aligned_points[:, 0] >= pca_x_min) & (aligned_points[:, 0] <=
                                              pca_x_max) &
        (aligned_points[:, 1] >= pca_y_min) & (aligned_points[:, 1] <=
                                              pca_y_max)
    )
    X_inside = aligned_points[inside_bbox_mask, 0]
    Y_inside = aligned_points[inside_bbox_mask, 1]
    Z_inside = aligned_points[inside_bbox_mask, 2]
    # Scatter plot for points inside the bounding box, colored by height
    im = ax.scatter(X_inside, Y_inside, c=-Z_inside, cmap='jet', s=1, alpha=0
                                          .8, label="Points Inside
                                          Bounding Box")
```

```
    # Draw bounding box
    ax.plot([pca_x_min, pca_x_max, pca_x_max, pca_x_min, pca_x_min],
            [pca_y_min, pca_y_min, pca_y_max, pca_y_max, pca_y_min],
            'w-', linewidth=2, label="Bounding Box")
    # Add colorbar for height
    fig.colorbar(im, ax=ax, label="Height (Z)")
    ax.set_xlabel("PCA X Coordinate")
    ax.set_ylabel("PCA Y Coordinate")
    ax.set_title("Original Point Cloud with bounding box on window")
    ax.legend()
    ax.axis("equal")
    # Save figure
    plt.savefig(os.path.join(output_dir, "bounding_box_original_height.png"))
    plt.close()

    # Figure 2 Interpolated Height Map
    print("Saving Figure 2: Interpolated Height Map of filtered points")
    fig, ax = plt.subplots(figsize=(7, 6))
    im2 = ax.imshow(grid_z_flipped, extent=(X_filtered.min(), X_filtered.max
                                          (), Y_filtered.min(),
                                          Y_filtered.max()),
                    origin="lower", cmap="jet")
    plt.colorbar(im2, ax=ax, label="Height (Z)")
    ax.set_xlabel("X Coordinate")
    ax.set_ylabel("Y Coordinate")
    ax.set_title(f"Interpolated Height Map of filtered points\nSurface
                                          Roughness (Ra) = {Ra:.6f} mm")
    plt.savefig(os.path.join(output_dir, "height_map_filtered_interpolated.
                                          png"))
    plt.close()

    #Figure 3 Raw height map
    print("Saving Figure 3: Raw Height Map of filtered points")
    fig, ax = plt.subplots(figsize=(7, 6))
    im2 = ax.scatter(X_filtered, Y_filtered, c=-Z_filtered, cmap="jet", s=1,
                                          alpha=0.8)
    plt.colorbar(im2, ax=ax, label="Height (Z)")
    ax.set_xlabel("X Coordinate")
    ax.set_ylabel("Y Coordinate")
    ax.set_title("Raw Filtered Point Cloud (Height-Based Coloring)")
    plt.savefig(os.path.join(output_dir, "height_map_filtered_raw.png"))
    plt.close()

# Function to calculate surface roughness
def calculate_surface_roughness(grid_z_flipped):
    print("Calculating surface roughness parameters (Ra, Rq, Rz)")
    mean_height = np.nanmean(grid_z_flipped)
    deviation_from_mean = np.abs(grid_z_flipped - mean_height)
    Ra = np.mean(deviation_from_mean)
    Rq = np.sqrt(np.mean(deviation_from_mean**2))
    Rz = np.nanmax(grid_z_flipped) - np.nanmin(grid_z_flipped)
    return Ra, Rq, Rz, mean_height

def average_z_in_range(Z_filtered, z_min, z_max):
    mask = (Z_filtered >= z_min) & (Z_filtered <= z_max)
    values_in_range = Z_filtered[mask]

    if len(values_in_range) == 0:
        print(f"No Z values found in range {z_min} to {z_max}")
        return np.nan

    avg_z = np.mean(values_in_range)
    print(f"Average Z in range [{z_min}, {z_max}] mm: {avg_z:.6f}")


# Main fucntion
def main(folder, name_data, name_figure, point_cloud_filtering=True):
    # Load and align point cloud
```

```
aligned_points = load_and_align_point_cloud(folder, name_data)

# Determin center and size of window
center_x = np.median(aligned_points[:, 0])
center_y = np.median(aligned_points[:, 1])
window_x = 10
window_y = 5

# Filter points
filtered_pca_xyz, filtered_mask, pca_x_min, pca_x_max, pca_y_min,
                                          pca_y_max =
                                          filter_points_by_pca_window(
                                          aligned_points, center_x,
                                          center_y, window_x, window_y)

# Filter point cloud
downsampled_points = filter_point_cloud(filtered_pca_xyz,
                                          point_cloud_filtering)

# Check if points remain
if len(downsampled_points) == 0:
    print("No valid points left after filtering and downsampling")
    return

# Create height map using the filtered points
X_filtered = downsampled_points[:, 0]
Y_filtered = downsampled_points[:, 1]
Z_filtered = downsampled_points[:, 2]

# Generate height map using grid interpolation
grid_x, grid_y = np.meshgrid(
    np.linspace(X_filtered.min(), X_filtered.max(), 500),
    np.linspace(Y_filtered.min(), Y_filtered.max(), 500)
)

grid_z = griddata((X_filtered, Y_filtered), Z_filtered, (grid_x, grid_y),
                                           method='linear')

# Handle null values
grid_z = np.nan_to_num(grid_z, nan=np.nanmedian(Z_filtered))

# Flip Z values by negating them so they are displayed correctly
grid_z_flipped = -grid_z

# Calculate surface roughness
Ra, Rq, Rz, mean_height = calculate_surface_roughness(grid_z_flipped)

# Save figures
output_dir = f"./code/data/{folder}/height_map_images/{name_data}"
os.makedirs(output_dir, exist_ok=True)

# Save height map figures
save_height_map_figures(X_filtered, Y_filtered, Z_filtered,
                                          grid_z_flipped, output_dir, Ra
                                          , aligned_points,
                                          filtered_pca_xyz, pca_x_min,
                                          pca_x_max, pca_y_min,
                                          pca_y_max)

# Save roughness calculations to a text file
roughness_file = os.path.join(output_dir, "roughness_calculations.txt")
with open(roughness_file, "w") as f:
    f.write(f"Surface Roughness Calculations for {name_data}:\n")
    f.write(f"Ra (Mean Roughness): {Ra:.6f} mm\n")
    f.write(f"Rq (Root Mean Square Roughness): {Rq:.6f} mm\n")
    f.write(f"Rz (Max Height Difference): {Rz:.6f} mm\n")
    f.write(f"Mean Height: {mean_height:.6f} mm\n")
    f.write(f"Min Height: {np.nanmin(grid_z_flipped):.6f} mm\n")
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            f.write(f"Max Height: {np.nanmax(grid_z_flipped):.6f} mm\n")
            f.write(f"Number of Points After Filtering: {len(downsampled_points)}
                                                        \n")

    # Generate Histogram
    print("Save histogram")
    fig, ax = plt.subplots(figsize=(7, 6))

    # Plot the histogram
    n, bins, patches = ax.hist(Z_filtered, bins=40, density=True, color='blue
                                            ', edgecolor='black', alpha=0.
                                            7)

    # Fit normal distribution
    mu, std = norm.fit(Z_filtered)

    # Make histogram
    fig, ax = plt.subplots(figsize=(7, 6))
    ax.hist(Z_filtered, bins=40, color='blue', edgecolor='black', alpha=0.7,
                                            density=True)

    # caluclate normal curve for histogram
    xmin, xmax = mu - 5*std, mu + 5*std
    x = np.linspace(xmin, xmax, 100)
    p = norm.pdf(x, mu, std)
    ax.plot(x, p, 'r--', linewidth=2, label=f'Normal Fit \n  ={mu:.4f},   ={
                                            std:.4f}')

    # Labels and title
    ax.set_xlabel("Height (mm)")
    ax.set_ylabel("Density")
    ax.set_title(f"Height Distribution Histogram of {name_figure}")
    ax.legend()

    plt.savefig(os.path.join(output_dir, "height_histogram.png"))
    plt.close()

    # Shapiro-Wilk test
    stat, p_value = shapiro(Z_filtered)
    print(f"Shapiro-Wilk Test: W={stat:.4f}, p-value={p_value:.4f}")
    if p_value > 0.05:
        print("Normally distributed, do not reject null hypoth")
    else:
        print("Not normally distributed, rejectt null hypot")

    print("Pipeline done")

# Run main function for single file, foldername, file_name, name of graphs
main("experiment_3/powderlayer_metalplate_v2", "63_pc.ply", "Mannitol powder
                                        layer", point_cloud_filtering=True
                                        )
```